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VLM-MSGraph: Vision Language Model-enabled Multi-hierarchical Scene Graph for robotic assembly

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ABSTRACT

Intelligent robotic assembly is becoming a pivotal component of the manufacturing sector, driven by growing demands for flexibility, sustainability, and resilience. Robots in manufacturing environments need perception, decision-making, and manipulation skills to support the flexible production of diverse products. However, traditional robotic assembly systems typically rely on time-consuming training processes specific to fixed settings, lacking generalization and zero-shot learning capabilities. To address these challenges, this paper introduces a Vision Language Model-enabled Multi-hierarchical Scene Graph (VLM-MSGraph) approach for robotic assembly, featuring generalized assembly sequence learning and 3D manipulation in open scenarios. The MSGraph incorporates high-level task planning structured as triplets, organized by multiple VLM agents. At a low level, the MSGraph retains 3D spatial relationships between industrial parts, enabling the robot to perform assembly tasks while accounting for object geometry for effective manipulation. Assembly drawings, physics simulations, and assembly tasks in a laboratory setting are used to evaluate the proposed system, advancing flexible automation in robotics.

1. Introduction

In today's transition to smart manufacturing, industrial companies are striving for flexible production of complex and personalized products [1]. Flexible automation aims to equip robots with proactive manipulation skills [2] to accommodate dynamic manufacturing requirements—such as sequential operations in assembly—overcoming the bottlenecks associated with pre-programmed procedures in large-scale production lines [3]. This requires robots to understand assembly tasks, decompose instructions into executable actions, and perform manipulations on diverse industrial components.

For task understanding, existing methods typically involve visually perceiving the surrounding workspace, followed by reasoning about the geometrical relationships of industrial parts for task allocation [4]. For task execution, robot manipulations are generally generated by imitating human demonstrations or explored through reinforcement learning-based planning policies [5]. Despite the advantages of these methods in task parsing, reasoning, and operation learning, they face significant challenges in generalization and transferability when applied to new assembly tasks.

Emerging Vision Language Models (VLMs) exhibit significant capabilities in few-shot reasoning and generalization for common-sense,

open-vocabulary tasks [6], showing great potential to enhance robot learning, interaction, and proactive execution abilities. However, during inference processes for manufacturing tasks, VLMs inevitably produce hallucinatory or erroneous content and lack domain-specific knowledge, rendering their reasoning results unsuitable for direct robot assembly applications. Furthermore, although VLMs can enhance robot task allocation abilities, industrial parts in assembly tasks often have unstructured shapes, making it difficult for robots to manipulate them without geometric reconstructions—such as grasp poses and 3D point clouds—let alone perform sequential operations.

To address these challenges, this paper proposes a robotic assembly method, VLM-MSGraph, which leverages VLMs to manage high-level assembly planning and to retain 3D object instances with geometric assembly states at a low level, creating a Multi-hierarchical Scene Graph (MSGraph) for the flexible assembly of mechanical products. First, the method enhances VLMs' inference capabilities in the industrial domain by using multiple agents and retrieval methods to learn assembly sequences, forming the high-level planning component of the MSGraph. Next, the MSGraph captures 3D assembly states of various products through RGB-D visual input from real-world environments. Finally, the robot proactively executes tasks by translating sequential assembly

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instructions from the MSGraph into actions with optimal object poses. This method simplifies product assembly configuration and enhances the generalization and adaptability of robotic operations in industrial tasks. The remainder of the paper is organized as follows. Section 2 reviews recent works related to flexible robotic automation, with a focus on core techniques. The VLM-MSGraph architecture, including its task planning algorithm and 3D memory methods, is presented in Section 3. Section 4 evaluates the performance of our flexible robotic assembly system on typical assembly tasks across various environmental settings. Finally, conclusions and future work are presented in Section 5.

2. Related works

This section begins with a recap of recent developments in flexible automation for smart manufacturing, followed by a review of advanced technologies, including VLMs and intelligent robot manipulation, that are driving this transformation.

2.1. Flexible automation for smart manufacturing

Flexible automation integrated with advanced information, communication, and analytics technologies allows manufacturing systems to easily adapt to changes in the types and quantities of products being produced, enhancing agility, efficiency, customization, and responsiveness to market demands [7]. Numerous manufacturing paradigms, such as self-organizing production [8], human–robot collaboration [9], and embodied AI [10], are emerging to facilitate flexible assembly of complex and personalized products in unstructured environments.

In diverse manufacturing applications, Giannetti and Essien [11] promoted predictive capabilities and fostered rapid decision-making abilities for smart factories to adapt to new circumstances. Wurster et al. [12] proposed condition-based control for hybrid disassembly systems by using a reinforcement learning model, solving flexible disassembly planning problems. Kahveci et al. [13] developed an assembly automation system for electric vehicle battery modules to support decision-making and improve both process and product quality with five interconnected layers of data acquisition, integration, storage, analytics, and visualization. Rozo et al. [14] presented a method for teaching and programming adaptive robots for the flexible assembly of e-bike motors by learning from demonstration. Although these systems demonstrate adaptability to some extent, their zero-shot and open-vocabulary capabilities warrant further exploration, particularly in the era of foundation models.

2.2. Vision language models in industrial settings

VLMs with more than one billion parameters, pre-trained on vast amounts of data, possess powerful reasoning and generalization abilities, leading to widespread applications. For example, the GPT series [15], LLaMA [16], LLaVA [17], and others are continuously being released to provide assistance in common scenarios for diverse tasks such as in-context learning and instruction following. Their surprising zero-shot reasoning performance enables VLMs to rapidly progress as a brain for interacting with the real world [18].

In industrial settings, two major advantages of VLMs are their generation and planning abilities. For instance, Huang et al. [19] utilized VLMs to generate robot trajectories for manipulation tasks by providing an open set of instructions and objects. Besides, Brohan et al. [20] proposed vision-language-action models to perform semantic reasoning for robot motion planning. Despite these advantages, in specific scenarios like manufacturing tasks, VLMs inevitably produce erroneous responses to human-imposed constraints. Therefore, the reasoning abilities of VLMs should be enhanced to incorporate domain knowledge via technologies like chain-of-thought [21], retrieval-augmented generation [22], or multiple-agent interaction [23]. Moreover, only appropriate VLM reasoning can effectively benefit robot manipulation in downstream tasks.

2.3. Intelligent robot manipulation

Nowadays, robot manipulation is becoming more intelligent through learning algorithms, enhancing generality and transferability across various tasks. For task planning, Jiang et al. [24] proposed action-conditioned 3D scene graphs, enabling robots to perform different actions in unstructured environments. For precise manipulation, Goyal et al. [25] introduced the RVT-2 model, which can learn multiple pegging and plugging operations from just a few demonstrations for new tasks. In general robot grasping tasks, Fang et al. [26] developed the AnyGrasp model, which generates accurate, 7-DoF, dense, and temporally smooth grasp poses for any object.

In manufacturing, Gkourmelos et al. [27] integrated digital twins with a natural language interface to enable behavior-based control of robots, facilitating intuitive interaction in assembly tasks. Wang et al. [28] and Liu et al. [29] explored vision-and-language-based robot navigation in manufacturing environments, where robots accurately interpreted human language instructions to complete tasks. However, intelligent robot manipulation skills, particularly those involving multiple operations in manufacturing tasks, remain in their infancy. Capabilities such as memory, decision-making, and actions for complex assemblies with geometrically constrained industrial parts are still underdeveloped. Motivated by this challenge, this work proposes the VLM-MSGraph for robot assembly, enabling task inference, 3D scene graph memory, and autonomous operations across diverse scenarios.

3. Methodology

The system architecture of the VLM-MSGraph is illustrated in Fig. 1. Given a visual input, such as onsite workspace settings or drawing images of a product, the VLMs analyze the required assembly sequences, infer simple sentences to describe the manipulations in each sequence, and generate triples to represent these actions. These triples form the high-level task planning within the MSGraph. Meanwhile, the MSGraph captures the 3D spatial relationships of the product's components in low-level memory through RGB-D visual observations. Specifically, the memory serves as 3D object instances in the MSGraph, where object classes, bounding boxes, and semantic masks are detected and projected into 3D point clouds through the perception module. Then, to enable the robot to execute task planning in the physical world, the optimal 6D pose for manipulating each component is estimated, allowing the robot to reconstruct the assembly process of the product based on the task planning and spatial relation memory embedded in the MSGraph.

3.1. Assembly sequence learning for MSGraph task planning

To minimize hallucinations in VLMs and align them with domain-specific knowledge in industrial settings, the assembly sequence learning incorporates chain-of-thought reasoning and retrieval-augmented generation strategies, as illustrated in Fig. 2. The method can infer assembly sequences from images, involving three key stages: (1) initial description of assembly steps and object relationships, (2) refined assembly steps with a designated assembly base, and (3) triple-check review for summary. The resulting assembly chain consists of sequentially ordered steps represented in concise triples.

Initial description of assembly steps and object relationships.

This stage focuses on identifying various elements in visual input across different image regions and determining both their spatial and assembly relationships. Specifically, to mitigate the hallucinatory issues in VLMs, a part list containing object names and quantities is used as retrieval information to match objects in the visual input during inference. The *Relation Narrator* agent describes the geometrical relationships (e.g., above, below, right, left, etc.) between different objects using the retrieved names and quantities, while the *Image Narrator* outlines the assembly steps (e.g., attach to, screw, go through, wrap,

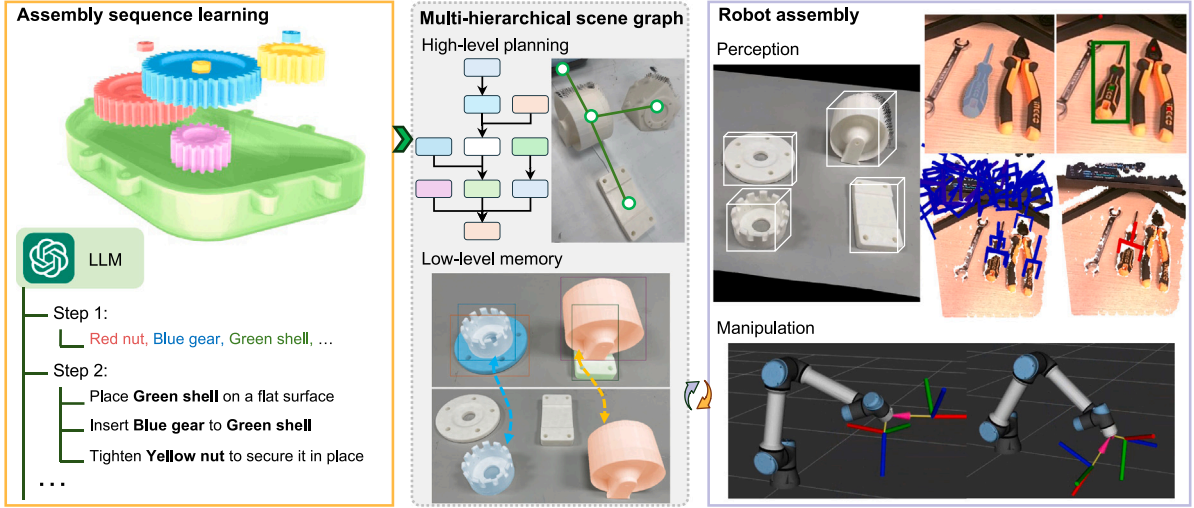


Fig. 1. The system architecture of VLM-MSGraph for robot assembly.

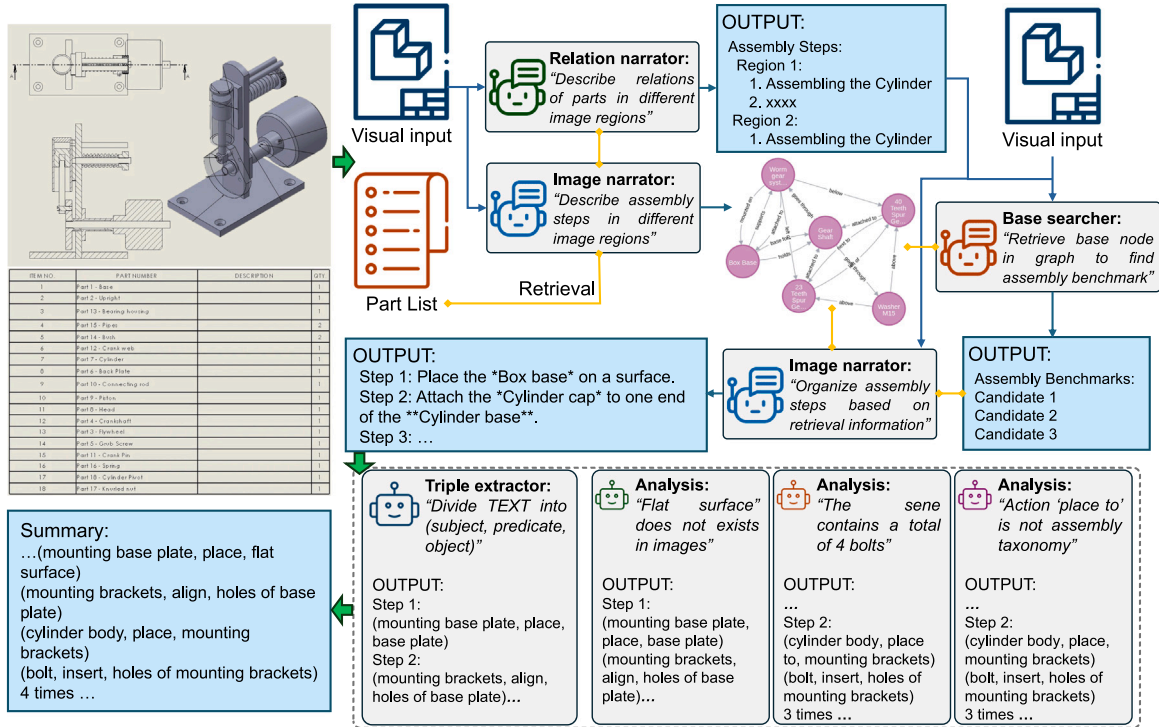


Fig. 2. Assembly sequence learning based on multiple AI agents and retrieval strategies.

etc.) in different image regions. These narrators produce descriptions in triples like $\langle \text{subject}, \text{predicate}, \text{object} \rangle$ based on GPT-4o. With the VLM agents, the narrator in stage 1 is represented by the following formula:

$$Nar_1^{VLM}(v, \epsilon, Prompt) = \bigcup_{i=1}^K \{ \hat{s}_i, \hat{p}_i, \hat{o}_i \} \quad (1)$$

where v , ϵ , and $Prompt$ represent visual input, examples in the part list, and relevant prompts, respectively. $\hat{s}_i, \hat{p}_i, \hat{o}_i$ denote the subject, predicate, and object in κ assembly steps.

Refined assembly steps with a designated assembly base. The output triples in stage 1 are translated into a graph G_{nar_1} to represent the assembly relationship chain of a product. The *Base Searcher* agent, acting as a retriever, accesses the graph G_{nar_1} and generates potential relational paths that map the relevant object connection information,

as denoted by:

$$Bsearch^{LLM}(Prompt, G_{nar_1}) \Rightarrow \{r_1, \rightarrow \dots, \rightarrow r_j\} \quad (2)$$

where r_j is the object presented in the visual input. Through continuous connections between edges and nodes, the *Base Searcher* agent retrieves paths ρ such as $\langle \text{cylinder cap} \rightarrow \text{attached to} \rightarrow \text{cylinder body} \rightarrow \dots \rightarrow \text{wooden plate} \rangle$ and identifies appropriate assembly benchmark σ to initiate the assembly process based on the graph and visual input. Next, another *Action Narrator* agent in stage 2 reorganizes the assembly steps starting from the identified benchmark σ for the entire assembly sequences, as represented by:

$$Nar_2^{VLM}(v, \rho, Prompt) = \bigcup \{ \hat{s}_i, \hat{p}_i, \hat{o}_i | i \in \{ \sigma \rightarrow \rho \} \} \quad (3)$$

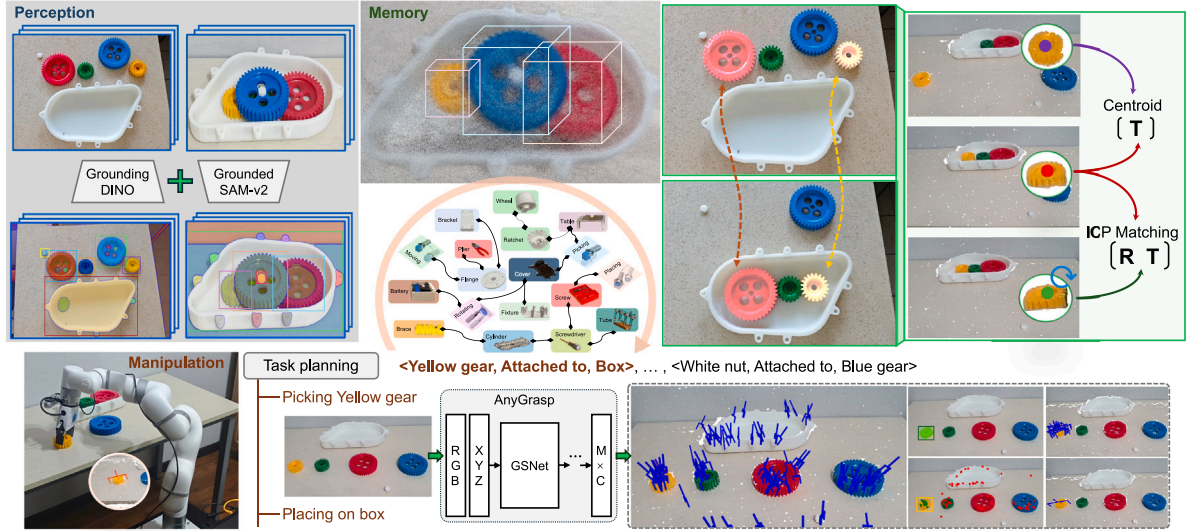


Fig. 3. Sequential robot assembly based on open-vocabulary perception, 3D object instance memory, and geometry-aware manipulation.

Triple-check review for summary. Based on the output from stage 2, the *Triple Extractor* agent first divides the assembly description into triples. At this stage, the results often contain biases, as VLM agents are not always accurate and can overlook predefined rules. For example, the output from the *Image Narrator* may indicate that the “Wingnut” is missing from the given drawing image, while “table”, “cover”, and “screw” are absent from the part list. Additionally, “grasp” might be incorrectly split into “grasp to”, misaligning with the rules set in the assembly relation. Therefore, this stage requires AI analysis to integrate drawing images v with the assembly relation and part list to identify and correct these omissions in the output from stage 2, while also providing necessary corrections. As depicted in Fig. 2, *Analysis 1* accurately identifies the absence of the “Wingnut” in the image and discovers the missing triple “(wingnut, insert, pipe)”. *Analysis 2*, adapting to the actual situation, adds additional triples like “(screw, fix, cylinder body)”, and *Analysis 3* correctly changes “grasp to” to “grasp”. Evidently, multiple agents can identify different issues in the output from the previous stage, and their analyses enhance the interpretability of these assembly steps while ensuring that all results conform to predefined rules, represented as triples consisting of a subject, an assembly relation, and an object. Lastly, the summarized triples are organized into the MSGraph $G_t = (N_t, E_t)$, representing the high-level task planning, where the nodes N_t represent the sets of objects in the triples and the edges E_t denote the action predicates between these objects in time step t .

3.2. 3D MSGraph memory for robot assembly

Aligned with high-level planning, the MSGraph captures spatial relationships for assembling various mechanical objects in physical settings. The robot autonomously executes sequential assembly operations using open-vocabulary perception, 3D object instance memory, and geometry-aware manipulation, as illustrated in Fig. 3.

Open-vocabulary perception. In the perception module, an RGB-D camera mounted on the robot’s end-effector captures sequential observations $O_t = [I_t^{rgb}, I_t^{depth}]$ over time t , where $I_t^{rgb} \in \mathbb{R}^{h \times w \times 3}$ represents an RGB image, and $I_t^{depth} \in \mathbb{R}^{h \times w}$ provides the corresponding depth map. Grounding DINO [30], which detects and identifies objects in images based on text prompts, outputs open-vocabulary detection labels and bounding boxes $c_i, i \in \{1, \dots, \eta\}$ for all mechanical parts in I_t . Additionally, GroundedSAM-v2 [31], which segments and tracks objects without labeled data, generates a set of object masks $m_i, i \in \{1, \dots, \eta\}$ by applying refined I_t within the bounding boxes. The detected labels

of each object’s bounding boxes and masks are then synchronized with nodes N recorded in the high-level MSGraph G .

3D object instance memory. To enable task planning within G , it is crucial to compute the geometric translation and rotation of the perceived objects between assembly and disassembly states, serving as manipulation memory. A 3D object instance is constructed to store the spatial relationships $P_i = \{R, T\}, i \in \{1, \dots, \eta\}$ at a low level, where rotation $R \in \mathbb{R}^{3 \times 3}$ is a 3×3 matrix in the special orthogonal group, and translation $T \in \mathbb{R}^3$ is a column vector representing rigid body motion. Within this framework, two types of information are retrieved from the visual input: ① observations of assembly states $O_s = \{O_s^1, O_s^2, \dots, O_s^\eta\}$, and ② observations of disassembly states $O_{dis} = \{O_{dis}^1, O_{dis}^2, \dots, O_{dis}^\eta\}$ for η mechanical parts, where each observation $O_t = \{I_1, I_2, \dots, I_k\}$ consists of RGB-D images captured from k camera views.

The 3D object instance $\zeta_i = \{p_i, c_i, m_i\}, i \in \{1, \dots, \eta\}$, comprising a 3D point cloud P_i and semantic feature m_i , represents nodes N in the low-level MSGraph G . For 3D point cloud aggregation, each pixel (u, v) in the depth map I_t^{depth} corresponding to a semantic feature m_i has a depth value $d_i(u, v)$. Using the camera intrinsic matrix K and the homogeneous coordinate vector p for the pixel (u, v) , the 3D point $P_i(u, v)$ is calculated as:

$$P_i(u, v) = d_i(u, v) \cdot K^{-1} p \quad (4)$$

The 3D point is then transformed into a common reference coordinate system using the camera pose $\Gamma_t \in \mathbb{R}^{4 \times 4}$:

$$P_i^w(u, v) = \Gamma_t \cdot P_i \quad (5)$$

where P_i is the homogeneous matrix. Subsequently, points from all views are aggregated to form the synchronized 3D point cloud for each semantic feature m_i ,

$$P_i = \bigcup_{j=1}^k \left\{ P_j^w(u, v) | I_j^{rgb}(u, v) \in m_i \right\} \quad (6)$$

The 3D object instances $\zeta_{i,s}$ in assembly states O_s and $\zeta_{i,dis}$ in disassembly states O_{dis} track the geometric and spatial changes of different mechanical parts as part of the low-level scene graph memory. First, a rough translation vector T_{rou} is calculated based on the distance between the centroids of each point cloud, $\zeta_{i,s}$ and $\zeta_{i,dis}$. Then, the Iterative Closest Point (ICP) algorithm [32] iteratively aligns the points in $\zeta_{i,dis}$ to $\zeta_{i,s}$ until convergence, producing a refined transformation $[R, T]$ that provides precise movement and rotation values for assembling mechanical parts based on 3D point clouds in G .

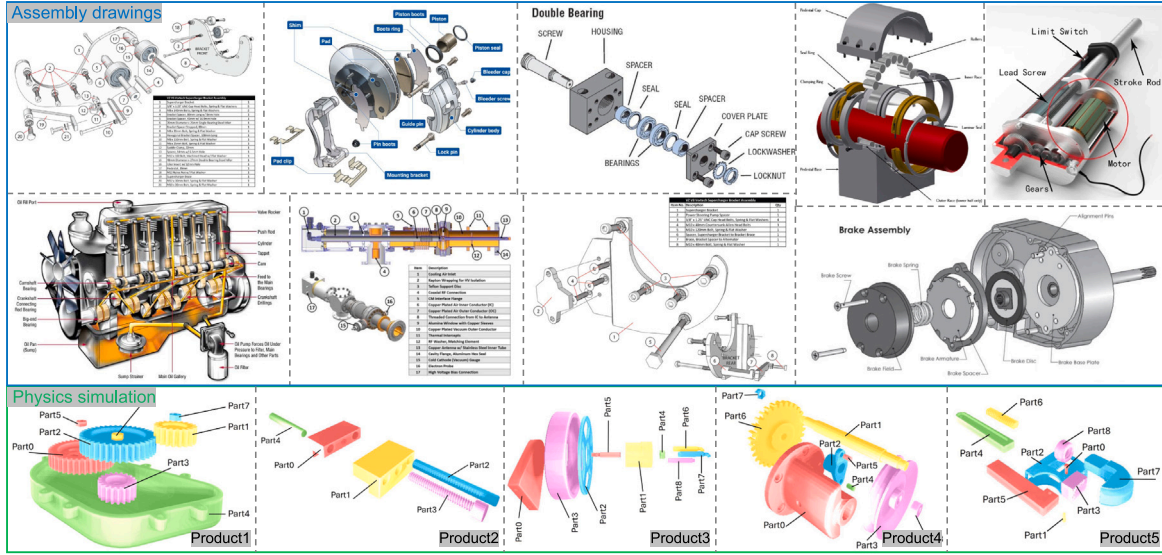


Fig. 4. Examples of assembly drawings and products in a physics simulation environment.

Geometry-aware manipulation. The robot manipulation module enables sequential assembly operations by integrating high-level task planning with spatial relationships stored in low-level memory from the MSGraph. To enable physical interaction with industrial parts, it is essential to estimate suitable grasp poses for the robot gripper. Since mechanical components often have unstructured shapes, they are challenging for a parallel-jaw gripper to handle. A pre-trained grasp generation module, AnyGrasp [26], is used to generate collision-free grasp poses for arbitrary industrial parts. Given an RGB image I_t^{rgb} and a depth map I_t^{depth} of the environment, the grasp module outputs potential grasp poses along with the transformation matrix $\Gamma_{any} \in \mathbb{R}^{4 \times 4}$. The grasp poses Γ_{any} include all observed objects, which can confuse the robot about which part to manipulate. To address this, the outputs of Grounding DINO c_i and GroundedSAM-v2 m_i serve as language queries to filter grasp poses for manipulation. Grasp poses from AnyGrasp are projected onto the c_i and m_i regions in the image, and those within the target object mask are selected as candidates for robot grasping, Γ_{grasp} . Using the transformation from the camera to the robot base Γ_b^c , obtained via hand-eye calibration, the world-coordinate grasp pose for a mechanical object is given by $\Gamma_{obj} = \Gamma_b^c \times \Gamma_{grasp}$. For each robot action E planned in high-level G , the corresponding low-level action is represented by $\bar{A}_t^i \in \mathbb{R}^7$, a 7-dimensional vector that includes the gripper's movement $\bar{T} = [d_t^x, d_t^y, d_t^z]$ in Cartesian space, derived from Γ_{obj} , the gripper's rotation $\bar{R} = [r_t^x, r_t^y, r_t^z]$ in Euler angles, also derived from Γ_{obj} , and a binary grasping state s_t indicating the open or close state of the gripper. Using these transformation matrices $[\bar{T}, \bar{R}]$, continuous trajectories for various robot actions are generated through inverse dynamics. For the initial and final parts of a trajectory, i.e., picking or placing an object, the robot gripper follows a trajectory to grasp or release the object required for assembly with the appropriate grasping state s_t :

$$\langle \bar{P}, \bar{P} \times \bar{R}, \bar{P} + 0.8\bar{T}, \bar{P} + 0.92\bar{T}, \bar{P} + 0.96\bar{T}, \bar{P} + \bar{T} \rangle \quad (7)$$

here, $\bar{P}$ represents the adjacent gripper location. The trajectory first adjusts the gripper's orientation to align with the grasp pose of the target object without altering its position. The gripper then approaches the object along a straight path with progressively smaller motions while maintaining its orientation. Once at the grasp position, the gripper closes to secure the object, ensuring a firm hold that prevents crushing or dropping.

4. Experiments

In this section, the performance of VLM-enabled assembly task planning is evaluated using both an assembly drawing dataset and a physics simulation environment. Subsequently, real-world robot assembly experiments, guided by the task planning, are conducted to demonstrate the effectiveness of the MSGraph in flexible production.

4.1. Evaluation of assembly task planning

Assembly drawing dataset. The assembly task planning for a product involves sequential assembly relationships. To efficiently evaluate the proposed method across diverse scenarios, an assembly drawing dataset for mechanical products was developed, containing 127 images, as shown in the upper part of Fig. 4. Key terms such as ball screws, bearings, bolts, brackets, brakes, centrifugal clutches, connectors, engines, gearboxes, motors, pneumatic cylinders, springs, and more were used in Google searches to compile the drawing images for the dataset. These selected products represent the most common connection types, including plug-in, pin, screw, key, buckle, and belt/cable connections. Additionally, the dataset is characterized by its generality and heterogeneity, presenting various types of assembly views.

The data was annotated using a hybrid approach that combined human and AI efforts. Initially, GPT-4o was employed to identify part names and quantities from the images and to generate preliminary assembly relationships between the identified industrial parts in sequential order. Human reviewers then reviewed and corrected any inaccuracies in the relationships and sequences proposed by GPT-4o. The final assembly task planning between different components of the product is expressed as sequential triplets in the format (*object name 1, assembly relation, object name 2*), as shown in the ground truth triplets in Fig. 6.

Multiple VLM agents, including GPT-4o, GPT-4omini, Llama3V, Llama3.1, and Llama-adapter, were used to compare assembly sequence learning performance on the assembly drawing dataset, as shown in Table 1. The prompt examples for VLM agents are illustrated in Fig. 5. It should be noted that additional prompts can be customized from these examples to apply assembly sequence learning across various scenarios. Besides, since Llama3.1 can only process language data, Llama3V and Llama-adapter2 were employed to extract language information from images. Metrics such as Recall (R) and mean Recall (mR) were used to evaluate the accuracy of predicted assembly task planning, where R measures the proportion of correctly identified assembly sequences for

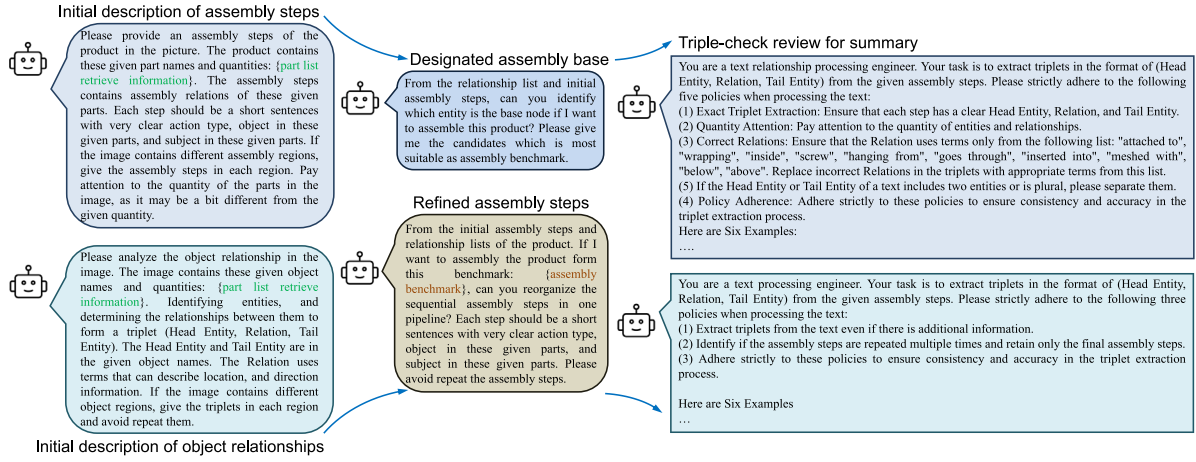


Fig. 5. Illustration of prompt examples for VLM agents.

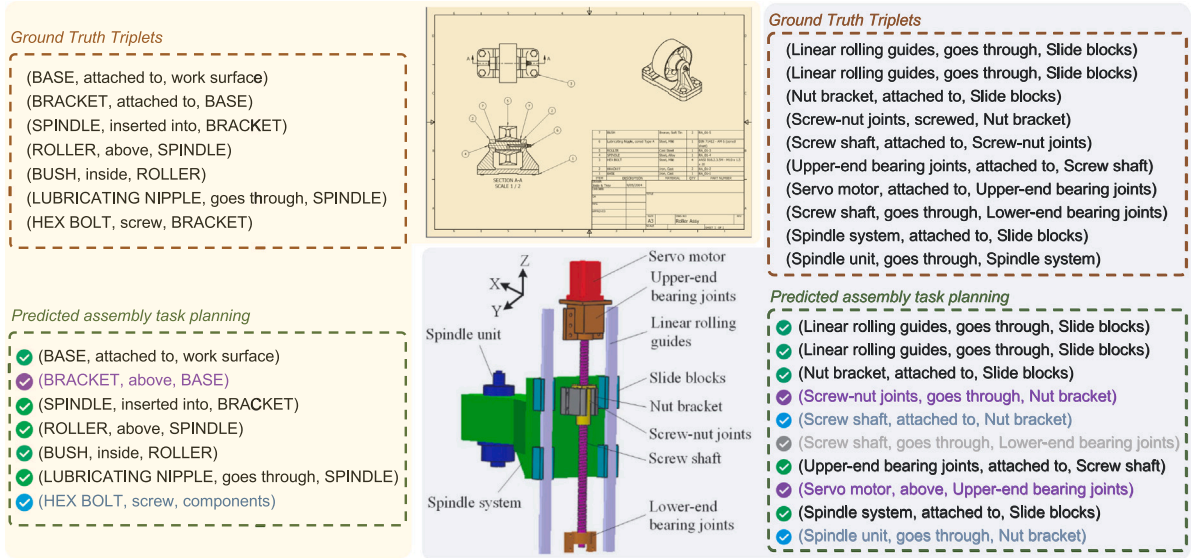


Fig. 6. Illustration of assembly task planning for assembly drawings.

Table 1
Results of assembly sequence learning on the assembly drawing dataset.

	R@Triplet	mR@Triplet	R@Pair	mR@Pair	R@Action	mR@Action
GPT-4o	30.25	28.35	58.03	62.49	58.11	59.95
GPT-4omini	22.85	19.44	53.10	56.90	48.51	51.76
Llama3V & Llama3.1	7.56	5.82	30.76	35.12	25.57	27.96
Llama-adapter2 & Llama3.1	3.79	3.09	12.23	12.32	8.75	10.17

a specific task, and mR averages these R values across all tasks to assess overall performance. The indices @Triplet, @Pair, and @Action indicate correct results when the entire triplet, object pair, and object-relation, respectively, are fully matched.

Overall, the accuracy performance of approximately 60% suggests that VLM agents can be aligned with the manufacturing domain and assembly knowledge through chain-of-thought reasoning and retrieval-augmented generation strategies. However, three main factors contribute to errors in assembly sequence learning for task planning. First, although VLM agents may generate assembly relations that are similar in describing an operation, they are marked as incorrect if they differ from the ground truth annotations, as shown in the purple hook results in Fig. 6. Second, in some cases, the VLM agents may confuse the subject and object in an assembly process, even though the incorrect output can still complete the task, as indicated by the

green hook results. Finally, there are often multiple valid assembly task plans for a product. However, reversed assembly sequences can reduce result accuracy due to rigid evaluation metrics for triples, even if the sequences are correct in real-world scenarios, as shown by the gray hook results in Fig. 6.

Physics simulation environment. The assembly drawing dataset is susceptible to human annotation errors, making it difficult to directly evaluate the accuracy of assembly task planning from the images. To overcome this, five products were selected for evaluation of their assembly procedures based on task planning learned by VLM agents within a physics simulation environment, DiffRedMax [33], as shown in the bottom part of Fig. 4. This physics-based simulator reliably simulates contact-rich assembly motions using a signed-distance-field representation, which supports complex collision geometries and shapes. For assembly path planning between different components, the breadth-first search algorithm [34] was employed to generate trajectories from

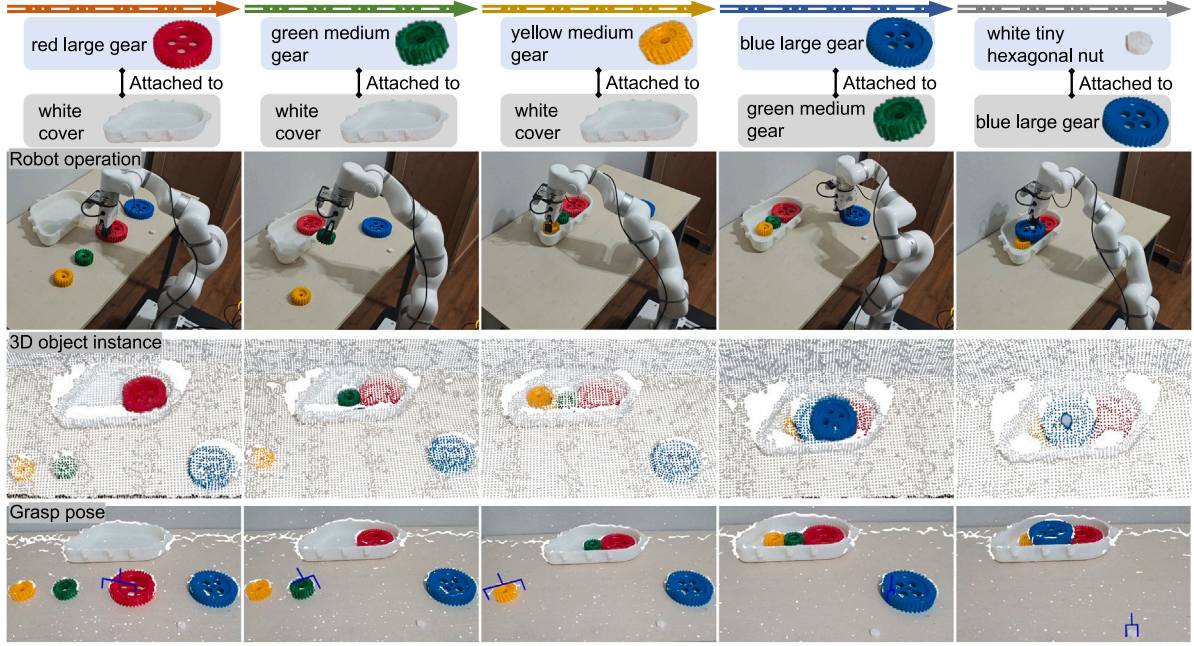


Fig. 7. Demonstration of robot assembly based on MSGraph memory and geometry-aware manipulation.

Table 2

Results of assembly tasks on the physics simulation environment.

Task	Without base	With base
Product1	12/20	17/20
Product2	18/20	20/20
Product3	19/20	20/20
Product4	13/20	18/20
Product5	7/20	15/20

disassembly states s_i to the final assembled state s_o , by iterating through the action space, which includes six translational and six rotational degrees of freedom.

The assembly results for these five products are presented in Table 2. The second row shows the number of successful experiments out of the total without the base searcher agent, while the third row shows the results with the base searcher agent. With assembly task planning learned by multiple VLM agents and path planning executed in the physics simulator, most products achieved a high success rate for their assembly processes.

4.2. Robot assembly based on MSGraph

Based on the learned task planning, it is crucial to evaluate the performance of robot assembly in a realistic environment.

Real-world assembly setup. A partial gearbox assembly is used to demonstrate open-vocabulary perception, 3D object instance memory, and geometry-aware manipulation. The desktop setup involves five assembly steps to attach red, green, yellow, and blue gears, as well as a white nut, onto a white cover. These tasks are performed by a UFACTORY xARM6 robotic arm mounted on an Agilex Ranger Mini 3 mobile base. A RealSense D455 camera serves as the visual sensor, enabling perception of the surrounding environment.

MSGraph Memory Results. Instead of relying on task-specific perception, open-vocabulary textual prompts are used to locate, classify, and segment objects from visual input. This is followed by mapping 3D object instance memory to retrieve geometric and spatial relations for robot assembly operations. As shown in Table 3, the textual prompts for assembly task planning from steps 1 to 5 are “red large gear”,

Table 3

Results of 3D object instance memory based on open-vocabulary perception.

Task	Textual prompt	Experiment
Step 1	red large gear	7/10
Step 2	green medium gear	9/10
Step 3	yellow medium gear	8/10
Step 4	blue large gear	7/10
Step 5	white tiny hexagonal nut	7/10

“green medium gear”, “yellow medium gear”, “blue large gear”, and “white tiny hexagonal nut”. These natural language prompts demonstrate the system’s generalization and transferability in perception. For the MSGraph memory, experimental results are provided in the last column of Table 3. In ten trials, over seven experiments in each task planning step successfully produced accurate 3D object instances and provided precise 3D movement and rotation values for robot assembly operations.

Demonstration of robot assembly process. The robot assembly steps from 1 to 5 are shown in Fig. 7. The first row illustrates high-level assembly task planning within MSGraph, detailing the manipulated objects, robot actions, and target objects, such as the triplet $\langle \text{red large gear, attached to, white cover} \rangle$. For task execution, the robot decomposes the action of “attached to” into picking the manipulated object “red large gear” and subsequently placing it onto the target “white cover”. The second row shows the robot performing the assembly task. The third row visualizes the 3D object instances, displaying 3D point clouds with distinct colors for semantic features. The 3D object instance memorizes the spatial relations of the object between assembly and disassembly states. The last row highlights suitable grasp poses for different mechanical parts based on their shapes, enabling geometry-aware manipulation to improve task stability and efficiency.

5. Conclusions

With the goal of creating configurable production lines that reduce cost, time, and the need for programming expertise, flexible automation has gained significant research interest in smart manufacturing systems, where robots must adaptively perform production tasks in dynamic

environments. To enable generalizable, open-set product assembly, this paper proposes VLM-MSGraph, which supports zero-shot assembly sequence learning and robot 3D manipulation without task specificity. In summary, the main scientific contributions of this research are twofold:

- (1) Proposed a VLM-enabled assembly sequence learning method to generate high-level task planning in MSGraph. This approach leverages multiple VLM agents and retrieval strategies to address hallucination and error generation issues commonly encountered with VLMs in the manufacturing domain. The method demonstrates broad applicability across diverse inputs, showing resilience to varying formats, perspectives, and product types in drawing images, facilitating contextual manufacturing knowledge learning across a range of domains.
- (2) Introduced an intelligent robot assembly based on 3D MSGraph low-level memory. The memory is characterized by open-vocabulary perception, semantic information, and 3D point clouds to enable geometry-aware manipulation for executing assembly task planning. This approach addresses challenges posed by unstructured mechanical part shapes in robot manipulation, paving the way for precise, dexterous robotic operations across diverse manufacturing tasks.

Moreover, based on experimental results in assembly drawings, physics simulations, and real-world environments, our proposed VLM-MSGraph method demonstrates effective performance in practical applications. Beyond these achievements, several promising future research directions are highlighted: (1) developing a closed-loop robot control program to handle subtle transformation deviations for precise assembly, and (2) extending desktop assembly tasks to a larger workspace with mobile robots for long-term navigation, interaction, and manipulation, facilitating a more flexible robotic system.

CRedit authorship contribution statement

Shufei Li: Writing – original draft, Validation, Methodology. **Zhi-jie Yan:** Visualization, Validation. **Zuoxu Wang:** Writing – review & editing, Supervision. **Yiping Gao:** Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Zuoxu Wang reports was provided by National Natural Science Foundation of China. Shufei Li reports was provided by State Key Laboratory of Intelligent Manufacturing Equipment and Technology. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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